Week 10: Advanced SQL Query System

The e-commerce company wants to perform advanced analysis on customer purchases, product pricing, and sales performance. The management team needs complex reports to identify high-value customers, expensive products, and business trends.

Find products whose price is greater than the average product price.

Step 1: Find average price

```
SQL> SELECT ROUND(AVG(PRICE), 2) AS AVERAGE_PRICE
  2  FROM PRODUCT;

AVERAGE_PRICE
-------------
        10400
```

Step 2: Compare products with average price

```
SQL> SELECT PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  WHERE PRICE > (
  4      SELECT AVG(PRICE)
  5      FROM PRODUCT
  6  )
  7  ORDER BY PRICE DESC;

PRODUCT_NAME                                         PRICE
-------------------------------------------------- ----------
Laptop                                                55000
Smartphone                                            28000
Tablet                                                22000
Gaming Monitor                                        18500
```

2. Find Products Above Average Price

The company wants to identify premium products.

1. Find products priced above the average price.

```
SQL> SELECT PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  WHERE PRICE > (
  4      SELECT AVG(PRICE)
  5      FROM PRODUCT
  6  )
  7  ORDER BY PRICE DESC;

PRODUCT_NAME                                         PRICE
-------------------------------------------------- ----------
Laptop                                                55000
Smartphone                                            28000
Tablet                                                22000
Gaming Monitor                                        18500
```

2. Display product name, category, and price.

```
SQL> SELECT
  2      P.PRODUCT_NAME,
  3      C.CATEGORY_NAME,
  4      P.PRICE
  5  FROM PRODUCT P
  6  JOIN CATEGORY C
  7      ON P.CATEGORY_ID = C.CATEGORY_ID
  8  WHERE P.PRICE > (
  9      SELECT AVG(PRICE)
 10      FROM PRODUCT
 11  )
 12  ORDER BY P.PRICE DESC;

PRODUCT_NAME
------------------------------------------------
CATEGORY_NAME                  PRICE
------------------------------ ----------
Laptop
Laptops                         55000

Smartphone
Mobile Devices                  28000

Tablet
Mobile Devices                  22000


PRODUCT_NAME
------------------------------------------------
CATEGORY_NAME                  PRICE
------------------------------ ----------
Gaming Monitor
Monitors                        18500
```

3. Identify expensive products in each category.

```
SQL> SELECT
  2      P.PRODUCT_NAME,
  3      C.CATEGORY_NAME,
  4      P.PRICE
  5  FROM PRODUCT P
  6  JOIN CATEGORY C
  7      ON P.CATEGORY_ID = C.CATEGORY_ID
  8  WHERE P.PRICE > (
  9      SELECT AVG(P2.PRICE)
 10      FROM PRODUCT P2
 11      WHERE P2.CATEGORY_ID = P.CATEGORY_ID
 12  )
 13  ORDER BY C.CATEGORY_NAME, P.PRICE DESC;

PRODUCT_NAME
------------------------------------------------
CATEGORY_NAME                  PRICE
------------------------------ ----------
Headphones
Audio Devices                    4200

Mechanical Keyboard
Computer Accessories             2500

Smartphone
Mobile Devices                  28000


PRODUCT_NAME
------------------------------------------------
CATEGORY_NAME                  PRICE
------------------------------ ----------
Power Bank
Power Accessories                1800
```

4. Find products with price higher than a selected category average.

```
SQL> SELECT
  2      PRODUCT_NAME,
  3      PRICE
  4  FROM PRODUCT
  5  WHERE CATEGORY_ID = (
  6      SELECT CATEGORY_ID
  7      FROM CATEGORY
  8      WHERE CATEGORY_NAME = 'Audio Devices'
  9  )
 10  AND PRICE > (
 11      SELECT AVG(P.PRICE)
 12      FROM PRODUCT P
 13      JOIN CATEGORY C
 14          ON P.CATEGORY_ID = C.CATEGORY_ID
 15      WHERE C.CATEGORY_NAME = 'Audio Devices'
 16  );

PRODUCT_NAME                                              PRICE
-------------------------------------------------- ----------
Headphones                                              4200
```

Query 1: Best-Selling Product

Find products with maximum sales quantity.

This query also displays revenue and includes ties.

```
SQL> SELECT
  2      P.PRODUCT_NAME,
  3      SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD,
  4      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  5  FROM PRODUCT P
  6  JOIN ORDERDETAIL OD
  7      ON P.PRODUCT_ID = OD.PRODUCTID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  HAVING SUM(OD.QUANTITY) = (
 10      SELECT MAX(TOTAL_QUANTITY)
 11      FROM (
 12          SELECT SUM(QUANTITY) AS TOTAL_QUANTITY
 13          FROM ORDERDETAIL
 14          GROUP BY PRODUCTID
 15      ) PRODUCT_SALES
 16  );

PRODUCT_NAME                                       TOTAL_QUANTITY_SOLD
-------------------------------------------------- -------------------
TOTAL_REVENUE
-------------
Smartphone                                                       6
       168000

USB Cable                                                        6
         2700
```

Query 2: High-Value Customers

Find customers whose total purchase amount is greater than average customer spending.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      SUM(O.TOTALAMOUNT) AS TOTAL_SPENDING
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME
  9  HAVING SUM(O.TOTALAMOUNT) > (
 10      SELECT AVG(CUSTOMER_TOTAL)
 11      FROM (
 12          SELECT SUM(TOTALAMOUNT) AS CUSTOMER_TOTAL
 13          FROM ORDERS
 14          GROUP BY CUSTOMERID
 15      ) CUSTOMER_SPENDING
 16  )
 17  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_I CUSTOMER_NAME                          TOTAL_SPENDING
---------- ---------------------------------------- --------------
CUST000001 Arjun Kumar                                       55850
CUST000003 Rahul Menon                                       28000
CUST000014 Lakshmi Venkatesan                                28000
CUST000005 Vikram Reddy                                      22000
CUST000009 Karthik Subramanian                               18500
```

Query 3: Category Performance Analysis

● Highest revenue-generating category.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
 10  HAVING SUM(OD.QUANTITY * OD.PRICE) = (
 11      SELECT MAX(CATEGORY_REVENUE)
 12      FROM (
 13          SELECT SUM(OD2.QUANTITY * OD2.PRICE) AS CATEGORY_REVENUE
 14          FROM PRODUCT P2
 15          JOIN ORDERDETAIL OD2
 16              ON P2.PRODUCT_ID = OD2.PRODUCTID
 17          GROUP BY P2.CATEGORY_ID
 18      ) CATEGORY_SALES
 19  );

CATEGORY_NAME                  TOTAL_REVENUE
------------------------------ -------------
Mobile Devices                        190000
```

● Number of products sold in each category.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      NVL(SUM(OD.QUANTITY), 0) AS TOTAL_PRODUCTS_SOLD
  4  FROM CATEGORY C
  5  LEFT JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  LEFT JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
 10  ORDER BY TOTAL_PRODUCTS_SOLD DESC;

CATEGORY_NAME                  TOTAL_PRODUCTS_SOLD
------------------------------ -------------------
Mobile Devices                                   7
Audio Devices                                    6
Power Accessories                                6
Storage Devices                                  2
Computer Accessories                             2
Wearables                                        2
Laptops                                          1
Monitors                                         1
Laptop Accessories                               1
Web Accessories                                  0
Home                                             0

CATEGORY_NAME                  TOTAL_PRODUCTS_SOLD
------------------------------ -------------------
Pharmacy                                         0
Jewellery                                        0
Clothing                                         0
Electronics                                      0
```

● Average product price by category.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      ROUND(AVG(P.PRICE), 2) AS AVERAGE_PRODUCT_PRICE
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
  8  ORDER BY AVERAGE_PRODUCT_PRICE DESC;

CATEGORY_NAME                  AVERAGE_PRODUCT_PRICE
------------------------------ ---------------------
Laptops                                        55000
Mobile Devices                                 25000
Monitors                                       18500
Wearables                                       6500
Storage Devices                                 5200
Audio Devices                                 3633.33
Web Accessories                                 2800
Computer Accessories                            1675
Laptop Accessories                              1500
Power Accessories                               1125

10 rows selected.
```

Query 4: Customer Purchase History

Display:

- Customer name
- Number of orders
- Total purchase amount
- Most purchased product

```
SQL> WITH CUSTOMER_ORDERS AS (
  2      SELECT
  3          CUSTOMERID,
  4          COUNT(*) AS NUMBER_OF_ORDERS,
  5          SUM(TOTALAMOUNT) AS TOTAL_PURCHASE_AMOUNT
  6      FROM ORDERS
  7      GROUP BY CUSTOMERID
  8  ),
  9  PRODUCT_PURCHASES AS (
 10      SELECT
 11          O.CUSTOMERID,
 12          P.PRODUCT_NAME,
 13          SUM(OD.QUANTITY) AS TOTAL_QUANTITY,
 14          ROW_NUMBER() OVER (
 15              PARTITION BY O.CUSTOMERID
 16              ORDER BY SUM(OD.QUANTITY) DESC, P.PRODUCT_NAME
 17          ) AS RN
 18      FROM ORDERS O
 19      JOIN ORDERDETAIL OD
 20          ON O.ORDERID = OD.ORDERID
 21      JOIN PRODUCT P
 22          ON OD.PRODUCTID = P.PRODUCT_ID
 23      GROUP BY O.CUSTOMERID, P.PRODUCT_ID, P.PRODUCT_NAME
 24  )
 25  SELECT
 26      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
 27      NVL(CO.NUMBER_OF_ORDERS, 0) AS NUMBER_OF_ORDERS,
 28      NVL(CO.TOTAL_PURCHASE_AMOUNT, 0) AS TOTAL_PURCHASE_AMOUNT,
 29      NVL(PP.PRODUCT_NAME, 'No Purchase') AS MOST_PURCHASED_PRODUCT
 30  FROM CUSTOMER C
 31  LEFT JOIN CUSTOMER_ORDERS CO
 32      ON C.CUSTOMER_ID = CO.CUSTOMERID
 33  LEFT JOIN PRODUCT_PURCHASES PP
 34      ON C.CUSTOMER_ID = PP.CUSTOMERID
 35      AND PP.RN = 1
 36  ORDER BY C.CUSTOMER_ID;
```

CUSTOMER_NAME	NUMBER_OF_ORDERS	TOTAL_PURCHASE_AMOUNT
MOST_PURCHASED_PRODUCT		
Arjun Kumar	1	55850
Laptop		
Priya Sharma	1	3350
Bluetooth Speaker		
Rahul Menon	1	28000
Smartphone		

CUSTOMER_NAME	NUMBER_OF_ORDERS	TOTAL_PURCHASE_AMOUNT
MOST_PURCHASED_PRODUCT		
Ananya Iyer	1	8300
USB Cable		
Vikram Reddy	1	22000
Tablet		
Meera Krishnan	1	7000
Headphones		

CUSTOMER_NAME	NUMBER_OF_ORDERS	TOTAL_PURCHASE_AMOUNT
MOST_PURCHASED_PRODUCT		
Suresh Nair	1	6500
Smart Watch		
Kavya Rajan	1	5700
External Hard Disk		
Karthik Subramanian	1	18500
Gaming Monitor		

CUSTOMER_NAME	NUMBER_OF_ORDERS	TOTAL_PURCHASE_AMOUNT
MOST_PURCHASED_PRODUCT		
Divya Pillai	1	5200
External Hard Disk		
Aditya Verma	1	7000
Headphones		
Sneha Patel	1	2450
Laptop Stand		

CUSTOMER_NAME	NUMBER_OF_ORDERS	TOTAL_PURCHASE_AMOUNT
MOST_PURCHASED_PRODUCT		
Rohan Bose	0	0
No Purchase		
Lakshmi Venkatesan	1	28000
No Purchase		
Nithin Murugan	1	8350
Wireless Earbuds		

15 rows selected.

5. Prepare Advanced SQL Report

Report 1: Premium Product Report

Display:

● Product Name

● Category

● Price

● Stock Availability

Condition:

Products above average price.

```
SQL> SELECT
  2      P.PRODUCT_NAME,
  3      C.CATEGORY_NAME,
  4      P.PRICE,
  5      I.STOCK_QUANTITY,
  6      I.STOCK_STATUS AS STOCK_AVAILABILITY
  7  FROM PRODUCT P
  8  JOIN CATEGORY C
  9      ON P.CATEGORY_ID = C.CATEGORY_ID
 10  LEFT JOIN INVENTORY I
 11      ON P.PRODUCT_ID = I.PRODUCT_ID
 12  WHERE P.PRICE > (
 13      SELECT AVG(PRICE)
 14      FROM PRODUCT
 15  )
 16  ORDER BY P.PRICE DESC;

PRODUCT_NAME
--------------------------------------------------
CATEGORY_NAME                  PRICE STOCK_QUANTITY STOCK_AVAIL
------------------------------ ----------- -------------- -----------
Laptop
Laptops                                55000             25 Available

Smartphone
Mobile Devices                         28000             35 Available

Tablet
Mobile Devices                         22000             30 Available


PRODUCT_NAME
--------------------------------------------------
CATEGORY_NAME                  PRICE STOCK_QUANTITY STOCK_AVAIL
------------------------------ ----------- -------------- -----------
Gaming Monitor
Monitors                               18500             20 Available
```

Report 2: Customer Value Report

Display:

● Customer Name

● Total Orders

● Total Spending

● Customer Category

Customer categories have been defined based on their total spending:

- Premium: spending above ₹50,000
- Regular: spending from ₹10,000 to ₹50,000
- Basic: spending below ₹10,000

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDERID) AS TOTAL_ORDERS,
  5      NVL(SUM(O.TOTALAMOUNT), 0) AS TOTAL_SPENDING,
  6      CASE
  7          WHEN NVL(SUM(O.TOTALAMOUNT), 0) > 50000
  8              THEN 'Premium'
  9          WHEN NVL(SUM(O.TOTALAMOUNT), 0) >= 10000
 10              THEN 'Regular'
 11          ELSE 'Basic'
 12      END AS CUSTOMER_CATEGORY
 13  FROM CUSTOMER C
 14  LEFT JOIN ORDERS O
 15      ON C.CUSTOMER_ID = O.CUSTOMERID
 16  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME
 17  ORDER BY TOTAL_SPENDING DESC;
```

```
CUSTOMER_I CUSTOMER_NAME                              TOTAL_ORDERS TOTAL_SPENDING
---------- ---------------------------------------- ------------ --------------
CUSTOME
-------
CUST000001 Arjun Kumar                                          1          55850
Premium

CUST000014 Lakshmi Venkatesan                                   1          28000
Regular

CUST000003 Rahul Menon                                          1          28000
Regular


CUSTOMER_I CUSTOMER_NAME                              TOTAL_ORDERS TOTAL_SPENDING
---------- ---------------------------------------- ------------ --------------
CUSTOME
-------
CUST000005 Vikram Reddy                                         1          22000
Regular

CUST000009 Karthik Subramanian                                  1          18500
Regular

CUST000015 Nithin Murugan                                       1           8350
Basic


CUSTOMER_I CUSTOMER_NAME                              TOTAL_ORDERS TOTAL_SPENDING
---------- ---------------------------------------- ------------ --------------
CUSTOME
-------
CUST000004 Ananya Iyer                                          1           8300
Basic

CUST000011 Aditya Verma                                         1           7000
Basic

CUST000006 Meera Krishnan                                       1           7000
Basic


CUSTOMER_I CUSTOMER_NAME                              TOTAL_ORDERS TOTAL_SPENDING
---------- ---------------------------------------- ------------ --------------
CUSTOME
-------
CUST000007 Suresh Nair                                          1           6500
Basic

CUST000008 Kavya Rajan                                          1           5700
Basic

CUST000010 Divya Pillai                                         1           5200
Basic


CUSTOMER_I CUSTOMER_NAME                              TOTAL_ORDERS TOTAL_SPENDING
---------- ---------------------------------------- ------------ --------------
CUSTOME
-------
CUST000002 Priya Sharma                                         1           3350
Basic

CUST000012 Sneha Patel                                          1           2450
Basic

CUST000013 Rohan Bose                                           0              0
Basic


15 rows selected.
```

Report 3: Sales Performance Report

Display:

● Product-wise sales

```
SQL> SELECT
  2      P.PRODUCT_NAME,
  3      SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  4      SUM(OD.QUANTITY * OD.PRICE) AS REVENUE_GENERATED
  5  FROM PRODUCT P
  6  JOIN ORDERDETAIL OD
  7      ON P.PRODUCT_ID = OD.PRODUCTID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  ORDER BY REVENUE_GENERATED DESC;
```

```
PRODUCT_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
Smartphone                                                     6
          168000

Laptop                                                         1
           55000

Tablet                                                         1
           22000


PRODUCT_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
Gaming Monitor                                                 1
           18500

Smart Watch                                                    2
           13000

Wireless Earbuds                                               3
           10500


PRODUCT_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
External Hard Disk                                             2
           10400

Headphones                                                     2
            8400

Bluetooth Speaker                                              1
            3200


PRODUCT_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
USB Cable                                                      6
            2700

Mechanical Keyboard                                            1
            2500

Laptop Stand                                                   1
            1500


PRODUCT_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
Wireless Mouse                                                 1
             850


13 rows selected.
```

● Category-wise revenue

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
 10  ORDER BY TOTAL_REVENUE DESC;

CATEGORY_NAME                  TOTAL_REVENUE
------------------------------ -------------
Mobile Devices                        190000
Laptops                                55000
Audio Devices                          22100
Monitors                               18500
Wearables                              13000
Storage Devices                        10400
Computer Accessories                    3350
Power Accessories                       2700
Laptop Accessories                      1500

9 rows selected.
```

● Top-selling products

Top 5 Selling Products based on the Quantity sold

```
SQL> SELECT
  2      P.PRODUCT_NAME,
  3      SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD
  4  FROM PRODUCT P
  5  JOIN ORDERDETAIL OD
  6      ON P.PRODUCT_ID = OD.PRODUCTID
  7  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  8  ORDER BY TOTAL_QUANTITY_SOLD DESC
  9  FETCH FIRST 5 ROWS ONLY;

PRODUCT_NAME                                       TOTAL_QUANTITY_SOLD
-------------------------------------------------- -------------------
Smartphone                                                           6
USB Cable                                                            6
Wireless Earbuds                                                     3
Smart Watch                                                          2
Headphones                                                           2
```

Report 4: Business Decision Report

● High-performing products

```
SQL> SELECT
  2      P.PRODUCT_NAME,
  3      SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  4      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  5  FROM PRODUCT P
  6  JOIN ORDERDETAIL OD
  7      ON P.PRODUCT_ID = OD.PRODUCTID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  HAVING SUM(OD.QUANTITY) > (
 10      SELECT AVG(PRODUCT_QUANTITY)
 11      FROM (
 12          SELECT SUM(QUANTITY) AS PRODUCT_QUANTITY
 13          FROM ORDERDETAIL
 14          GROUP BY PRODUCTID
 15      ) SALES
 16  )
 17  ORDER BY QUANTITY_SOLD DESC;

PRODUCT_NAME                                       QUANTITY_SOLD TOTAL_REVENUE
-------------------------------------------------- ------------- -------------
Smartphone                                                     6        168000
USB Cable                                                      6          2700
Wireless Earbuds                                               3         10500
```

● Low-performing products

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      P.STOCK_QUANTITY,
  5      NVL(SUM(OD.QUANTITY), 0) AS QUANTITY_SOLD
  6  FROM PRODUCT P
  7  LEFT JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME, P.STOCK_QUANTITY
 10  HAVING NVL(SUM(OD.QUANTITY), 0) <= 1
 11  ORDER BY QUANTITY_SOLD;

PRODUCT PRODUCT_NAME                                  STOCK_QUANTITY
------- -------------------------------------------------- --------------
QUANTITY_SOLD
-------------
P000007 Power Bank                                                    75
              0

P000011 Web Camera                                                    50
              0

P000003 Mechanical Keyboard                                           60
              1


PRODUCT PRODUCT_NAME                                  STOCK_QUANTITY
------- -------------------------------------------------- --------------
QUANTITY_SOLD
-------------
P000004 Bluetooth Speaker                                             45
              1

P000014 Laptop Stand                                                  80
              1

P000010 Tablet                                                        30
              1


PRODUCT PRODUCT_NAME                                  STOCK_QUANTITY
------- -------------------------------------------------- --------------
QUANTITY_SOLD
-------------
P000013 Gaming Monitor                                                20
              1

P000002 Wireless Mouse                                               100
              1

P000001 Laptop                                                        25
              1


9 rows selected.
```

● High-value customers

```
SQL> SELECT
  2      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3      SUM(O.TOTALAMOUNT) AS TOTAL_SPENDING
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6      ON C.CUSTOMER_ID = O.CUSTOMERID
  7  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME
  8  HAVING SUM(O.TOTALAMOUNT) > 20000
  9  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_NAME                              TOTAL_SPENDING
---------------------------------------- --------------
Arjun Kumar                                         55850
Lakshmi Venkatesan                                  28000
Rahul Menon                                         28000
Vikram Reddy                                        22000
```

● Revenue trends

```
SQL> SELECT
  2      TO_CHAR(ORDERDATE, 'YYYY-MM') AS SALES_MONTH,
  3      COUNT(ORDERID) AS NUMBER_OF_ORDERS,
  4      SUM(TOTALAMOUNT) AS MONTHLY_REVENUE
  5  FROM ORDERS
  6  GROUP BY TO_CHAR(ORDERDATE, 'YYYY-MM')
  7  ORDER BY SALES_MONTH;

SALES_M NUMBER_OF_ORDERS MONTHLY_REVENUE
------- ---------------- ---------------
2026-01                2           59200
2026-02                2           36300
2026-03                2           29000
2026-04                2           12200
2026-05                2           23700
2026-06                2            9450
2026-08                1           28000
2026-09                1            8350

8 rows selected.
```